COMPANION RESEARCH GUIDE

Deontological ethics in AI alignment

A curated annotated bibliography for *The Deontological Architecture of Artificial Intelligence: Principles, Alignment, and Benchmarking*

Executive summary

The report's strongest evidence forms a five-part chain: philosophical definitions clarify duties and rights; deontic logic exposes formal conflicts; constitutional methods operationalize explicit principles; benchmarks test rule recognition and exceptions; and the EU AI Act translates rights-based constraints into governance obligations.

This bibliography selects twelve sources from that chain and adds a practical use case plus a critical limitation for each. The central caution is consistent across the literature: explicit rules improve auditability, but neither formal consistency nor benchmark accuracy proves moral correctness.

Scope. Curated core bibliography, not a line-by-line annotation of all 56 links in the source report. Priority is given to canonical philosophy, primary research, benchmark validation, and binding governance.

CORE SOURCES

Philosophical foundations

Selected for direct relevance to the report's argument and for practical value in AI alignment research.

01

PHILOSOPHICAL FOUNDATIONS / ORIENTATION

Alexander, Larry, and Michael Moore. "Deontological Ethics." Stanford Encyclopedia of Philosophy, Summer 2026 archive.

https://plato.stanford.edu/archives/sum2026/entries/ethics-deontological/

Contribution A rigorous map of agent-centered and patient-centered deontology, contractualism, thresholds, and the familiar objections to rule-based ethics. It is the best single source for separating distinct deontological claims that the report sometimes treats together.

Use in the report Use this entry to define the report's taxonomy and to frame the central design choice: whether an AI is constrained because of the agent's duties, the affected person's rights, or principles no one could reasonably reject.

Critical note This is conceptual synthesis, not an implementation guide. Its categories do not determine how rules should be represented, ranked, or tested in a deployed model.

02

PHILOSOPHICAL FOUNDATIONS / PRIMARY FRAMEWORK

Johnson, Robert, and Adam Cureton. "Kant's Moral Philosophy." Stanford Encyclopedia of Philosophy.

https://plato.stanford.edu/entries/kant-moral/

Contribution Explains the categorical imperative, universal law, humanity as an end, autonomy, and the relation between duty and moral worth. These concepts anchor the report's account of non-instrumental treatment and procedural respect.

Use in the report Use it to distinguish a Kantian constraint from a generic safety rule. A rule is Kantian only when its justification turns on universalizability, rational agency, dignity, or reciprocal legislation - not merely on desirable outcomes.

Critical note The entry concerns human moral agents. Applying its claims to machine behavior requires an explicit bridge between simulated rule compliance and the autonomy Kant thought necessary for moral agency.

CORE SOURCES

From plural duties to formal norms

Selected for direct relevance to the report's argument and for practical value in AI alignment research.

03

PHILOSOPHICAL FOUNDATIONS / CONFLICTING DUTIES

"William David Ross." Stanford Encyclopedia of Philosophy, Spring 2026 archive.

https://plato.stanford.edu/archives/spr2026/entries/william-david-ross/

Contribution Presents Ross's pluralistic account of prima facie duties and the role of judgment when fidelity, justice, beneficence, and non-injury conflict. It supports the report's move away from one-rule absolutism.

Use in the report Use Ross to motivate architectures that surface multiple applicable duties before adjudication. The framework is especially useful for explaining why a safety system may need both hard constraints and a transparent conflict-resolution layer.

Critical note Ross offers no mechanical weighting rule. Turning practical judgment into scores or priorities risks importing a consequentialist optimizer under deontological labels.

04

FORMALIZATION / LOGIC

McNamara, Paul, and Frederik Van De Putte. "Deontic Logic." Stanford Encyclopedia of Philosophy, Winter 2025 edition.

https://plato.stanford.edu/entries/logic-deontic/

Contribution Surveys formal systems for obligation, permission, prohibition, conditional norms, contrary-to-duty obligations, and normative conflict. It supplies the logical vocabulary behind the report's discussion of Chisholm-style paradoxes.

Use in the report Use it when specifying rule engines, policy checkers, or formal verification. The entry explains why a model needs context-sensitive and conditional norms rather than a flat list of obligations.

Critical note Formal consistency is not moral correctness. A deontic system can be internally valid while encoding the wrong facts, priorities, stakeholders, or exceptions.

CORE SOURCES

From theorem proving to constitutional training

Selected for direct relevance to the report's argument and for practical value in AI alignment research.

05 FORMALIZATION / THEOREM PROVING

Singh, Lavanya. "Automated Kantian Ethics: A Faithful Implementation." German Conference on Artificial Intelligence, 2022, pp. 187-208.

http://arxiv.org/pdf/2207.10152

Contribution Formalizes Kantian reasoning in dyadic deontic logic and implements it in Isabelle/HOL. The work is a concrete demonstration that philosophically informed ethical rules can be represented, tested, and explained through theorem proving.

Use in the report Use it as evidence for auditable neuro-symbolic layers: a generative model proposes an action or maxim, while a formal system checks permissibility and records the proof path.

Critical note The implementation is a research prototype. Its explainability applies to the encoded formal model; it does not guarantee that real-world situations are translated into that model accurately.

06 ALIGNMENT ARCHITECTURES / CONSTITUTIONAL TRAINING

Bai, Yuntao, et al. "Constitutional AI: Harmlessness from AI Feedback." arXiv:2212.08073, 2022.

https://arxiv.org/abs/2212.08073

Contribution Introduces a two-stage method in which a model critiques and revises outputs against written principles, then learns from AI-generated preferences. It is the report's clearest operational example of explicit rules shaping model behavior at scale.

Use in the report Use it to show how natural-language principles can become training signals rather than only post-generation filters. It also makes value choices more inspectable than a large pool of unstructured preference labels.

Critical note The method is not purely deontological: the reinforcement-learning stage still optimizes a learned preference signal. Compliance also depends on the model's interpretation of the constitution and the coverage of training prompts.

CORE SOURCES

Who writes the rules - and what counts as fair?

Selected for direct relevance to the report's argument and for practical value in AI alignment research.

---

07 ALIGNMENT ARCHITECTURES / PUBLIC INPUT

Huang, Saffron, et al. "Collective Constitutional AI: Aligning a Language Model with Public Input." FAccT 2024, pp. 1395-1417.

http://arxiv.org/abs/2406.07814

Contribution Extends constitutional training with a process for sourcing principles from a target public and comparing the resulting model with a developer-authored baseline. It turns the origin of rules into an explicit governance question.

Use in the report Use it to support the report's social-contract strand and to design a traceable pipeline from participant selection through principle synthesis, model training, and evaluation.

Critical note Participation does not automatically establish legitimacy. Sampling, facilitation, aggregation, and transformation of public input can all reintroduce institutional bias.

---

08 ALIGNMENT ARCHITECTURES / PROCEDURAL FAIRNESS

Mougan, Carlos, and Joshua Brand. "Kantian Deontology Meets AI Alignment: Towards Morally Grounded Fairness Metrics." arXiv:2311.05227, rev. 2024.

https://arxiv.org/html/2311.05227v2

Contribution Argues that outcome-parity metrics should be supplemented by Kantian attention to procedure, dignity, and non-instrumental treatment. It supports the report's claim that fair results can still arise from objectionable processes.

Use in the report Use it when evaluating systems that meet aggregate group metrics but alter individual decisions merely to satisfy a target. It provides a bridge from moral philosophy to fairness design criteria.

Critical note The paper is primarily conceptual. It does not yet supply a validated, domain-ready metric suite or resolve trade-offs between procedural constraints and outcome disparities.

CORE SOURCES

Benchmarks need both scores and validity

Selected for direct relevance to the report's argument and for practical value in AI alignment research.

09 BENCHMARKS / BASELINE EVALUATION

Hendrycks, Dan, et al. "Aligning AI With Shared Human Values." International Conference on Learning Representations, 2021.

https://arxiv.org/html/2008.02275v6

Contribution Introduces the ETHICS benchmark, spanning commonsense morality, deontology, justice, virtue ethics, and utilitarianism. Its deontology task gives the report a concrete baseline for testing recognition of role-based duties and constraints.

Use in the report Use it for comparative evaluation across models and training methods, but report results at the subtask level. Pair accuracy with adversarial and paraphrase tests to probe whether models rely on shortcuts.

Critical note A correct label does not establish deontological reasoning. The benchmark measures performance on textual judgments, not reliable action, generalization, or a defensible moral theory.

10 BENCHMARKS / VALIDITY AUDIT

Hancox-Li, Leif, and Borhane Blili-Hamelin. "Is ETHICS about ethics? Evaluating the ETHICS Benchmark." NeurIPS EvalEval Workshop, 2024.

https://arxiv.org/abs/2410.13009v2

Contribution Interrogates whether ETHICS subtasks actually measure the theories named by the benchmark. It is essential corrective reading because rule-following can be compatible with several moral theories and may not isolate deontological structure.

Use in the report Use it to define construct-validity checks before claiming a model is more ethical or more deontological. Evaluation should test reasons, counterfactual consistency, and rule selection - not only final labels.

Critical note The paper is a focused benchmark critique rather than a replacement evaluation framework. Teams still need domain-specific tasks and behavioral tests.

CORE SOURCES

Exceptions and governance

Selected for direct relevance to the report's argument and for practical value in AI alignment research.

11 BENCHMARKS / EXCEPTIONS

Jin, Zhijing, et al. "When to Make Exceptions: Exploring Language Models as Accounts of Human Moral Judgment." NeurIPS 35, 2022, pp. 28458-28473.

https://nips.cc/virtual/2022/poster/55320

Contribution Introduces MoralExceptQA and a structured Moral Chain of Thought approach for cases in which a normally binding rule may be overridden. It directly addresses the rigidity problem highlighted throughout the report.

Use in the report Use it alongside ETHICS: one task probes rule recognition, the other probes justified exception handling. Record the stated rule, its purpose, affected parties, severity, and the reason an exception is or is not permitted.

Critical note The dataset models human judgments from a limited population and should not be treated as universal moral truth. Reasoning traces can also sound coherent without being causally responsible for the answer.

12 GOVERNANCE / BINDING LAW

European Parliament and Council of the European Union. Regulation (EU) 2024/1689 (Artificial Intelligence Act). Official Journal of the European Union, 12 July 2024.

https://www.europarl.europa.eu/news/cs/press-room/20231206IPR15699/artificial-intelligence-act-deal-on-comprehensive-rules-for-trustworthy-ai

Contribution Creates a risk-based legal framework with categorical prohibitions, obligations for high-risk systems, transparency duties, and governance mechanisms. The prohibited-practices layer is the report's strongest policy analogue for rights-based constraints.

Use in the report Use the Act as a compliance boundary and source of auditable obligations, especially for prohibited and high-risk uses. Keep a crosswalk from each legal requirement to system controls, evidence, owners, and monitoring.

Critical note A legal risk taxonomy is not a full moral theory, and the linked page is a legislative overview rather than the controlling text. Teams should consult the current Official Journal text and counsel for implementation.

SYNTHESIS

A reading order that moves from theory to deployment

Read in sequence when designing or evaluating a deontological alignment program.

1. **Define the moral object**
   Start with Alexander and Moore, Kant, and Ross. Decide whether the system protects duties, rights, reciprocal principles, or a plural set of prima facie obligations.

2. **Specify conflict behavior**
   Use deontic logic to represent obligation, permission, prohibition, conditional rules, and contrary-to-duty recovery. Document which norms outrank others and why.

3. **Choose the enforcement layer**
   Compare theorem-proving checks with constitutional training. The former offers formal proofs over a model; the latter shapes model behavior but retains statistical uncertainty.

4. **Legitimate the constitution**
   Use Collective Constitutional AI to examine who selected the principles. Add stakeholder analysis, dissent handling, versioning, and an appeal mechanism.

5. **Test rules and exceptions separately**
   Pair ETHICS with MoralExceptQA, then apply Hancox-Li and Blili-Hamelin's validity critique. Require explanation consistency, counterfactual tests, and adversarial paraphrases.

6. **Map ethics to governance**
   Crosswalk the EU AI Act's prohibitions and high-risk duties to concrete controls. Keep legal compliance distinct from the broader moral case for a system.

> **Design implication.** A robust deontological stack is layered: principle provenance -> formal policy -> model behavior -> benchmark evidence -> deployment controls -> human appeal.

Selection method. The twelve entries were chosen from the report's works cited for authority, direct relevance, and coverage of the report's main claims. Secondary summaries, commercial commentary, and duplicate landing pages were deprioritized.